Práctica dos: Sistema Cardiovascular.

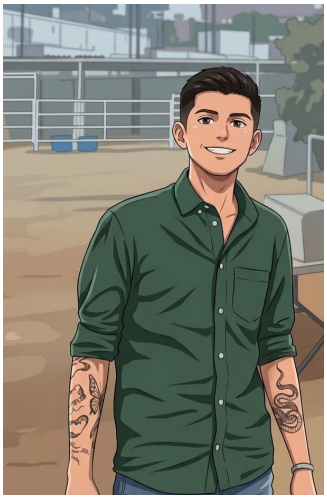
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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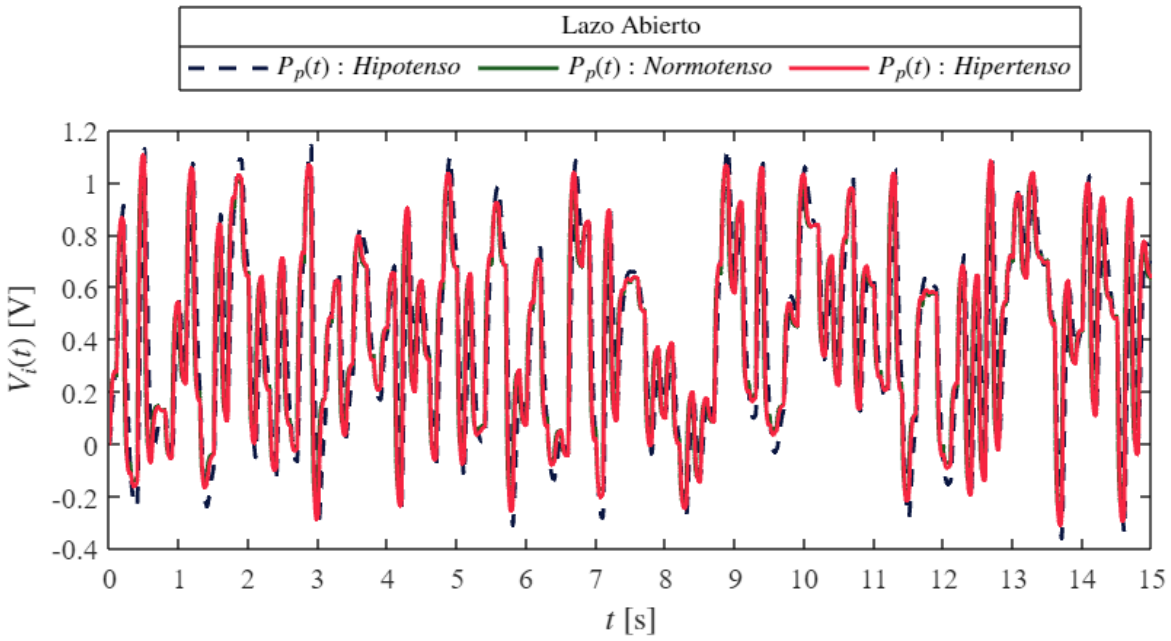
Datos de la simulación

```
clc; clear; close all; warning('off','all')
```

```
open_system('Sistema')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x = sim('Sistema', parameters);
t_data = (0:0.1:15)';
t = (0:0.01:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(Pa,'signal.xlsx');
```

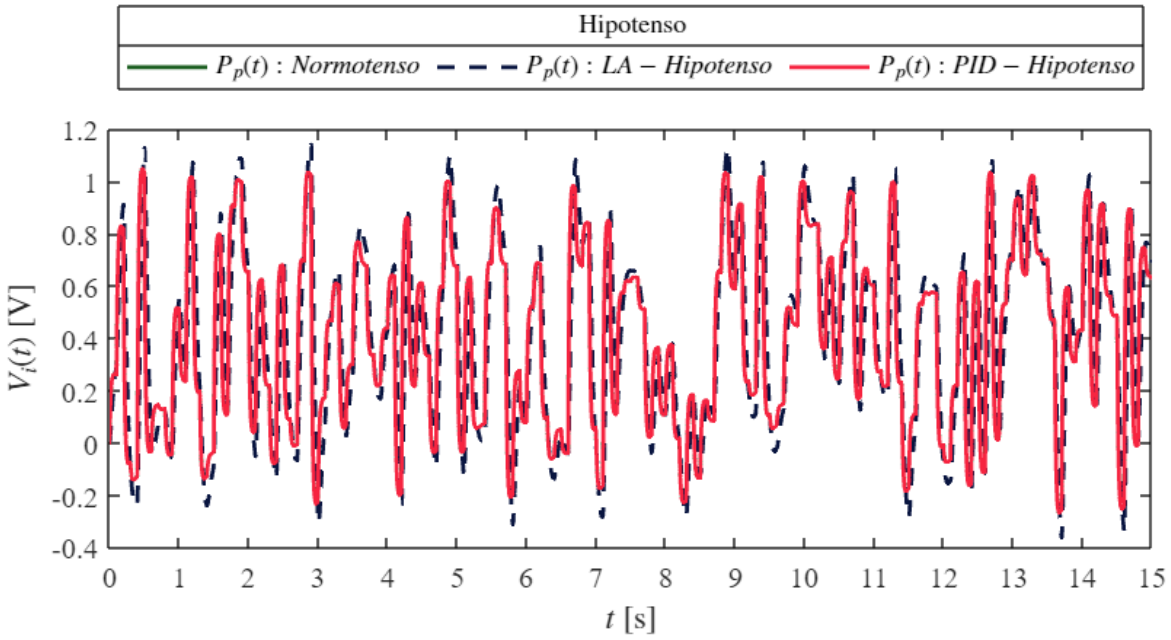
Resultados: Normotenso, Hipotenso e Hipertenso.

```
plotsignals(x.t, x.PpL, x.PpN, x.PpH,'Lazo Abierto')
```



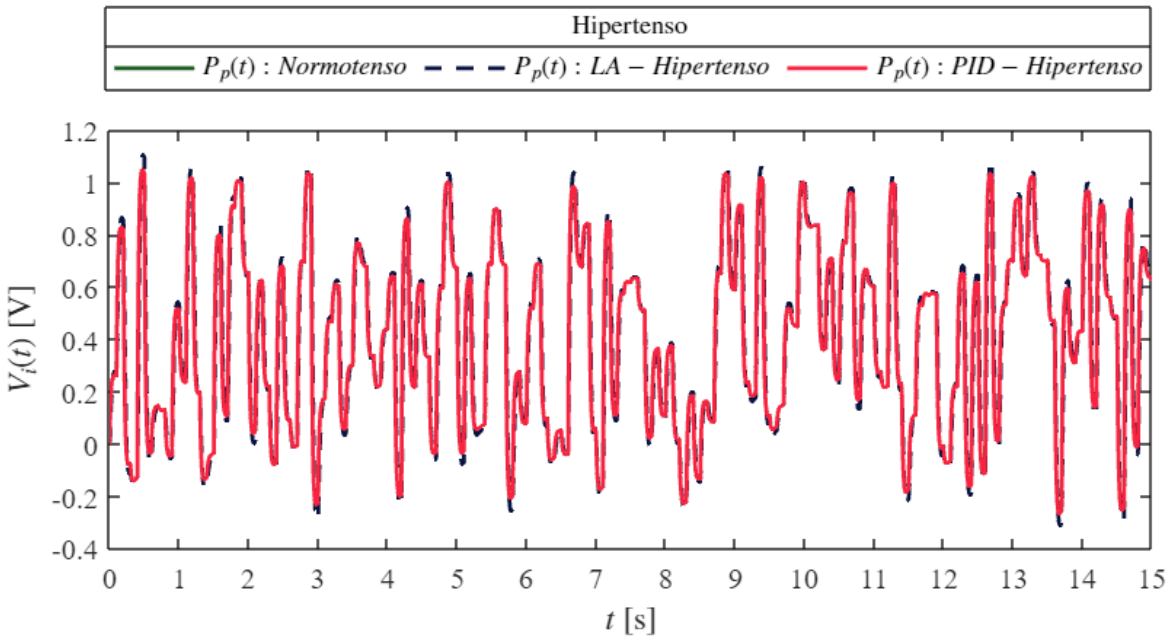
Resultados: Normotenso, Hipotenso y PID Hipotenso.

```
plotsignals1(x.t, x.PpN, x.PpL, x.PIDL,'Hipotenso')
```



Resultados: Normotenso, Hipertenso y PID Hipertenso.

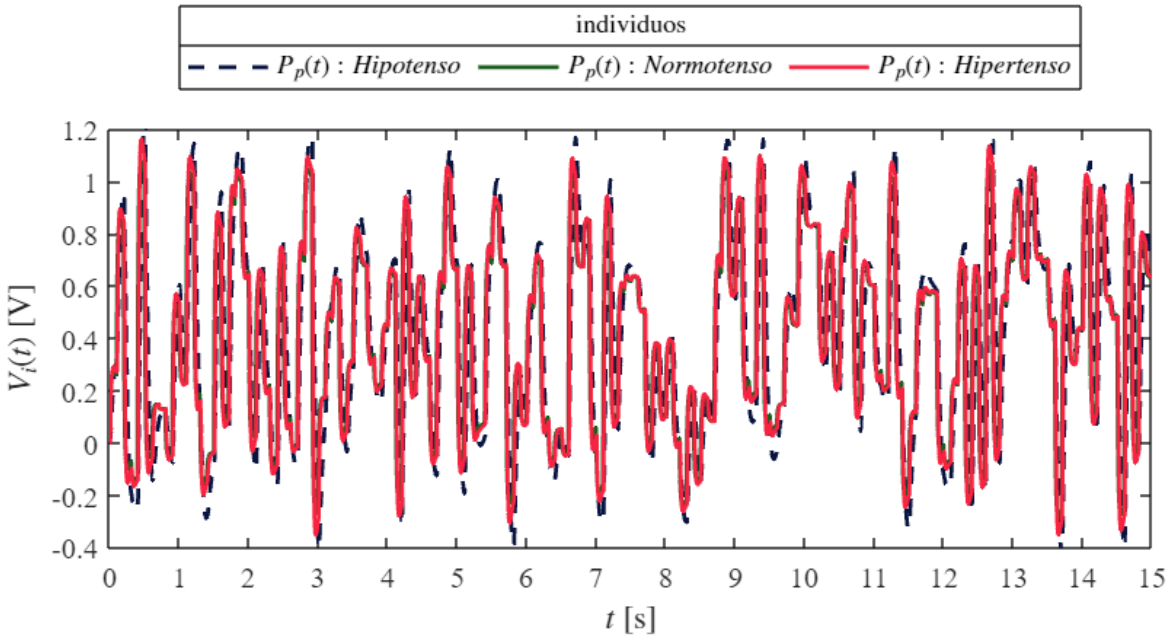
```
plotsignals2(x.t, x.PpN, x.PpH, x.PIDH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.010,0.001,1.000,3.500]; %[Z,L,R,C]
par_normo = [0.010,0.001,1.000,1.500];
par_hiper = [0.010,0.001,1.000,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
```

```
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'individuos')
```




```
function Pp = cardio(par,Pa,t)
    Z= par(1); L= par(2); R= par(3); C= par(4);
    dt = t(2);
    n = round(t(end)/dt);

    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;

    for i = 1:n
        FL(i+1) = FL(i) + (-(1/L)*Pp(i) + (1/L)*Pa(i))*dt;
        Pp(i+1) = Pp(i) + ((1/C)*FL(i) - (1/(Z*C) + 1/(R*C))*Pp(i) + 1/
(Z*C)*Pa(i))*dt;
    end
end
```

Función: Respuestas de la simulación

```
function plotsignals(t,x1,x2,x3,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [9,21,64]/255;
    clr2 = [27,94,32]/255;
    clr3 = [246,36,64]/255;
    order = [clr1;clr2;clr3]; colororder(order)
```

```
plot(t,x1,'--',t,x2,'-',t,x3,'-','lineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));

    L = legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');

set(L,'interpreter','latex','location','northoutside','orientation','Horizontal')
    title(L,[signal])

    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]','Interpreter','latex')

    xlim([0 15]); xticks(0:1:15)

    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'contenttype','vector')
end

function plotsignals1(t,x1,x2,x3,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [27,94,32]/255;
    clr2 = [9,21,64]/255;
    clr3 = [246,36,64]/255;
    order = [clr1;clr2;clr3]; colororder(order)


plot(t,x1,'-',t,x2,'--',t,x3,'-','lineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));

    L = legend('$P_p(t):Normotenso$','$P_p(t):LA-Hipotenso$','$P_p(t):PID-
Hipotenso$');

set(L,'interpreter','latex','location','northoutside','orientation','Horizontal')
    title(L,[signal])

    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]','Interpreter','latex')

    xlim([0 15]); xticks(0:1:15)

    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'contenttype','vector')
end
```

```
function plotsignals2(t,x1,x2,x3,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [27,94,32]/255;
    clr2 = [9,21,64]/255;
    clr3 = [246,36,64]/255;
    order = [clr1;clr2;clr3]; colororder(order)


plot(t,x1,'-',t,x2,'--',t,x3,'-','lineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));

    L = legend('$P_p(t):Normotenso$','$P_p(t):LA-Hipertenso$','$P_p(t):PID-
Hipertenso$');

set(L,'interpreter','latex','location','northoutside','orientation','Horizontal')
    title(L,[signal])

    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]','Interpreter','latex')

    xlim([0 15]); xticks(0:1:15)

    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'contenttype','vector')
end
```